

Anthony Ibarra
Embedded Systems Engineer
(Local to Chicago, IL; Available to Start Onsite)
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Summary:

Electrical and Computer Engineering graduate looking for the opportunity to become a professional in my respective field. Showcasing a strong focus in Embedded Systems, Control system theory, Robotics and Automation. With fundamental understanding of pattern recognition, Signal processing, and power electronics making me a flexible Embedded & Control Systems Engineer. Possessing various knowledge in different fields providing fast adaptable capability within the work environment of a multidisciplinary team. With the eagerness to learn on and off the job, demonstrating reliability and trustworthiness for a great addition toward a multidisciplinary team.

Education:

University of Illinois Chicago (UIC)

Jan 2023 - May 2025

Master of Science in Electrical and Computer Engineering Chicago, United States

Relevant Coursework:

- Audio & Acoustic Signal Processing, Filter Synthesis, Adaptive Digital Filters, Electromechanical Energy Conversion
- Linear Systems Theory & Design, Electromagnetic Compatibility, Advanced Computer Communication
- Convex Optimization, Mechatronics Embedded Design, Neural Networks

University of Illinois Chicago (UIC)

Aug 2017 - May 2022

Bachelor of Science in Computer Engineering Chicago, United States

Relevant Coursework:

- Artificial Intelligence, Principles of Modern & Auto Control, Pattern Recognition
- Computer Comm Networks, Robotics: Algorithm/Control, Embedded Systems

Certifications:

- Creating and Securing Bluetooth Low Energy (BLE) Application - LinkedIn Learning
- Advanced 5G NR (New Radio) Technologies - LinkedIn Learning
- Satellite Internet and Communications: Fundamentals - LinkedIn Learning
- C Programming for Embedded Applications - LinkedIn Learning
- Learning FPGA Development - LinkedIn Learning
- Learning Verilog for FPGA Development - LinkedIn Learning
- Introduction to FreeRTOS and Basic Task Management - LinkedIn Learning
- Getting Started with RISC-V - LinkedIn Learning
- Learning LabVIEW - LinkedIn Learning
- Data Acquisition with LabVIEW - LinkedIn Learning
- Robot Framework Test Automation: Level 1 (Selenium) - LinkedIn Learning

Skills:

- Altium, ANSYS HFSS, LTspice, PCB, soldering, SolidWorks
- HIL, MATLAB Simulink, Quartus, Icarus Verilog
- Raspberry PI, Arduino IDE, assembly language,
- C, C++, Python, Bash, FPGA Verilog, VHDL
- BLE, UDP, TCP/IP, I2C, UART, SPI, Ethernet, RF, LTE
- Wireshark, Ignition, Rockwell, PLC, Robotics, ROS2, SCADA, HMI, drivers,
- Git, GitHub, Docker, IEEE, OOP, IoT, Ubuntu, MySQL
- Latex, Microsoft PowerPoint Microsoft Word,
- Linux, Ubuntu, Visual Studio Code, VNC

Experience:

University of Illinois Chicago (UIC) Chicago, United States

Jan 2025 – May 2025

Research Project, Acoustic & Audio Signal Processing

- Expanded and executed a solution to remove unnecessary dampening from motorcycle helmets, ensuring clear sound quality for riders with hearing impairments while minimizing unwanted noise.
- This feature improves rider awareness in densely populated areas, potentially enhancing overall safety while traveling. The report was structured in the scope and style of a Signal Processing Society conference paper.
- Designed and enforced a specific method or solution, which was successfully applied to an existing problem.

- Conduct real-world testing and validation of project and research implementations, achieving improvements in performance or outcomes.
- Put in practice advanced algorithms in acoustic and audio signal processing, driving achievement of the project goal and delivering measurable results.

University of Illinois Chicago (UIC) Chicago, United States
Optimization on Three Coil Long Range (WPT)

Jan 2024 - May 2024

- Electromagnetic Compatibility testing using a spectrum analyzer, Oscilloscopes for EMC and EMI testing for PCB and signal integrity using RF SMA coaxial cables, SMA terminators
- Applied HFSS CAD and simulated a three-coil configuration for long-range wireless power transfer, validating findings from research papers.
- Developed and analyzed a mathematically simulated plot in MATLAB for three coils as demonstrated in the research paper, enabling quantitative evaluation of system performance across various parameter configurations.
- Conducted simulations and generated plots using Ansys HFSS 3D EM software to analyze and compare three-coil behavior, leading to data-driven insights for system optimization.
- Authored a professionally typeset IEEE-formatted report using LaTeX, clearly presented discoveries and results, and improved documentation quality.

University of Illinois Chicago (UIC) Chicago, United States
Self-Driving Car, Mechatronics Embedded Design

Jan 2023 - May 2023

- Directed a cross-functional team to manage time-sensitive development and design projects, implementing strategic decisions that ensured on-time completion of both product and hardware deliverables.
- Implemented real-time embedded software designs for sensors and encoders, developed filters for line camera or velocity input measurements, and tested and debugged microcontroller software to ensure system reliability.
- Created a motor driver circuit using FET Driver implementation—including single FET, half bridge, or full H-bridge configurations—controlled via a PWM input signal from the microcontroller.
- Designed a Boost Converter for the implementation of multiple power systems.
- Carried out product testing using oscilloscopes, multimeters, and benchtop power supplies, and performed soldering repairs as needed, resulting in improved quality control and measurable enhancement in defect reduction.
- Develop and tune PID controllers for both steering and velocity controls (Servo motor, DC motor) respectively.
- Design, develop the PCB through Altium Designer and get it manufactured, providing all necessary files such as BOM, gerber files, etc.
- Soldered all through-hole and SMT components, employed masking for PCB and prototype assemblies, and created necessary jumper wires through crimping and soldering.
- Prototyped and implemented a perf board circuit that served as an emergency backup in the event of PCB failure, ensuring uninterrupted continuation of product development.

University of Illinois Chicago (UIC) Chicago, United States
Senior Design Product Development

Aug 2021 - May 2022

- Collaborated in a four-member team with minimal supervision; to prototype an automated plant watering system that utilized moisture levels, outdoor weather conditions, and plant-specific data for efficient irrigation.
- Manage team through verbal and written communication to make sure all assignments and goals are completed on time, and submit weekly reports based on progress of project development
- Developed embedded programming for an Arduino Nano IoT 33 to optimize watering decisions based on plants' moisture levels and water consumption.
- Developed a mobile app using the Kivy framework to present essential information about the system, plant moisture levels, and weather conditions.
- Developed a UDP client-server connection to enable communication between the powered product and the mobile application, which provided manual control over the wireless connection.
- Implemented wireless communication using Bluetooth Low Energy (BLE) protocol to enable seamless connectivity between consumers and their devices.

Languages:

- Spanish (B-1)
- English (Advanced)